

NANNING, China

WMO: 594310

Lat: 22.7840N

Lon: 108.5472E

Elev: 503

StdP: 14.43

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	40.6	42.9	28.2	22.6	54.8	32.2	27.1	52.2	15.4	65.7	13.6	61.7	3.7	340	0.321

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	95.1	79.6	93.5	79.3	91.8	78.9	82.1	90.1	81.4	88.9	80.7	87.8	5.9	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.3	160.4	86.2	79.2	154.5	84.9	78.9	152.9	84.5	46.7	90.6	45.8	89.0	44.9	88.5	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.4	12.3	10.8	35.5	99.0	2.6	1.4	33.6	100.0	32.0	100.9	30.6	101.6	28.7	102.7		
(5)			34.6	83.9	2.6	0.9	32.7	84.5	31.2	85.0	29.8	85.5	27.9	86.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	72.3	55.9	59.4	64.8	73.8	79.6	83.0	83.8	83.6	81.0	75.4	67.4	58.8	(6)
(7)		DBStd	11.59	7.69	8.81	7.69	6.68	4.47	3.05	2.81	2.77	3.76	4.90	6.19	6.49	(7)
(8)		HDD50	44	25	12	1	0	0	0	0	0	0	0	0	6	(8)
(9)		HDD65	880	300	201	105	13	0	0	0	0	0	2	45	214	(9)
(10)		CDD50	8172	207	276	459	715	919	989	1047	1041	929	788	523	279	(10)
(11)		CDD65	3535	17	45	98	279	454	539	582	576	479	326	118	22	(11)
(12)		CDH74	33725	43	207	467	2016	4237	5862	6547	6353	4743	2474	693	83	(12)
(13)		CDH80	12703	2	47	109	636	1569	2276	2663	2611	1857	797	130	6	(13)
(14)	Wind	WSAvg	4.5	4.6	4.8	4.7	5.0	4.8	4.5	4.7	4.1	4.2	4.1	4.0	4.3	(14)
(15)	Precipitation	PrecAvg	65.3	1.9	1.5	2.6	3.5	7.1	12.2	13.7	9.9	6.0	3.0	2.4	1.4	(15)
(16)		PrecMax	93.6	6.8	6.1	6.7	10.4	14.8	21.1	26.1	18.9	13.2	9.1	8.8	4.8	(16)
(17)		PrecMin	49.8	0.2	0.2	0.3	0.7	3.3	2.9	5.5	3.1	0.9	0.2	0.1	0.1	(17)
(18)		PrecStd	10.4	1.7	1.3	1.5	2.3	2.5	4.3	5.4	3.5	2.9	2.5	2.2	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.5	86.0	87.8	93.1	95.3	95.4	96.9	96.7	95.1	91.6	86.3	80.4	(19)
(20)			MCWB	67.8	70.9	74.3	78.3	80.2	80.4	79.9	79.5	78.1	74.2	73.3	69.8	(20)
(21)		2%	DB	74.8	79.7	82.7	89.5	92.4	93.5	94.9	94.8	93.0	89.3	83.7	76.2	(21)
(22)			MCWB	65.4	69.4	73.1	76.8	79.0	80.2	79.9	79.4	77.7	73.9	71.0	66.5	(22)
(23)		5%	DB	71.4	75.5	79.0	86.4	90.0	91.6	93.0	93.0	91.2	87.0	80.9	73.2	(23)
(24)			MCWB	64.0	67.0	71.0	75.6	77.6	79.7	79.8	79.4	77.2	73.2	68.9	63.9	(24)
(25)		10%	DB	68.1	71.8	75.4	83.9	88.0	89.7	91.0	91.1	89.3	84.5	78.4	69.9	(25)
(26)			MCWB	62.2	66.0	69.4	74.4	77.1	79.2	79.7	79.3	76.7	72.0	67.8	62.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.7	72.4	75.7	79.7	82.4	83.3	83.3	82.7	81.0	78.4	76.2	72.8	(27)
(28)			MCDWB	75.7	82.0	84.5	89.3	92.0	92.0	92.2	90.8	88.3	85.6	81.8	78.0	(28)
(29)		2%	WB	67.4	70.5	73.8	78.1	80.8	81.9	81.9	81.6	80.1	77.1	74.5	69.8	(29)
(30)			MCDWB	72.2	77.7	81.1	86.7	89.0	89.4	89.8	89.3	87.2	84.0	79.9	73.9	(30)
(31)		5%	WB	65.6	68.7	72.3	76.8	79.8	81.0	81.0	80.8	79.4	76.1	72.8	67.1	(31)
(32)			MCDWB	69.7	74.1	77.9	84.4	87.0	88.0	88.2	88.1	86.3	82.6	78.0	70.7	(32)
(33)		10%	WB	63.6	66.9	70.6	75.5	78.8	80.2	80.2	80.1	78.6	75.0	71.1	63.8	(33)
(34)			MCDWB	67.4	71.0	74.8	81.9	85.1	86.5	86.8	86.9	85.2	81.1	75.8	68.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	12.4	12.4	11.8	13.3	13.9	12.9	13.0	13.8	14.8	15.7	16.1	15.3	(35)
(36)			MCDBR	18.4	19.0	18.1	18.1	17.5	16.1	16.9	17.1	17.8	18.9	19.8	20.3	(36)
(37)			MCWBR	9.9	9.5	9.2	7.7	6.7	5.6	5.1	5.4	5.6	6.4	8.7	10.4	(37)
(38)		5% WB	MCDWB	14.9	17.4	16.5	16.4	15.1	13.8	14.3	14.9	15.2	15.2	15.0	15.2	(38)
(39)			MCWBR	8.9	9.7	9.5	7.7	6.6	5.6	5.3	5.6	5.5	6.2	8.2	9.7	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.616	0.644	0.792	0.751	0.575	0.549	0.522	0.591	0.635	0.721	0.608	0.627	(40)
(41)		taud		1.675	1.655	1.425	1.541	1.956	2.046	2.115	1.904	1.769	1.561	1.744	1.674	(41)
(42)		Ebn at Noon		204	211	188	200	238	242	249	233	219	190	205	195	(42)
(43)		Edh at Noon		71	77	101	91	60	54	51	62	70	83	66	69	(43)
(44)	All-Sky Solar Radiation	RadAvg		675	766	839	1105	1329	1338	1468	1499	1420	1235	982	790	(44)
(45)		RadStd		167	144	122	134	124	174	142	90	92	101	155	149	(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	+0.50	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	+0.81	N/A	N/A	N/A	N/A	N/A	+112

Nomenclature: See separate page